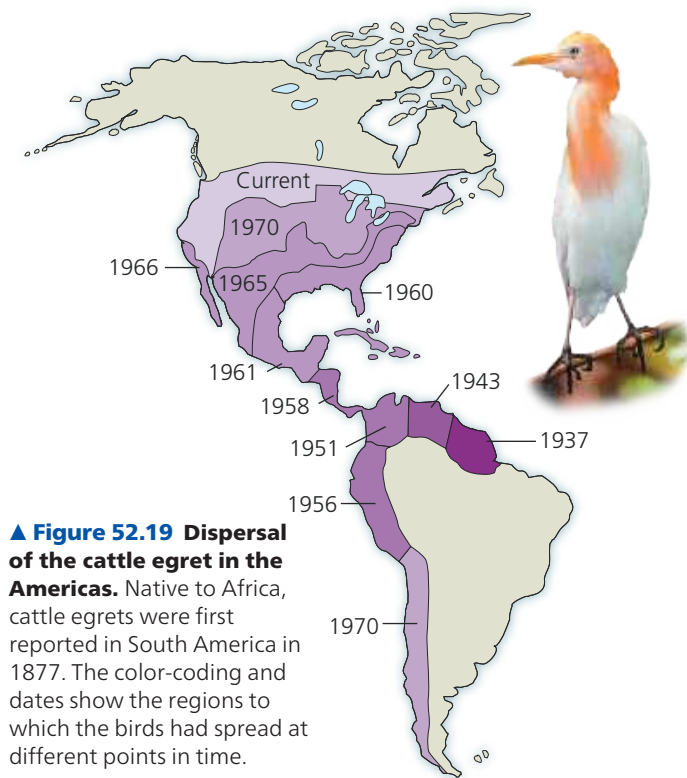




▲ **Figure 52.19 Dispersal of the cattle egret in the Americas.** Native to Africa, cattle egrets were first reported in South America in 1877. The color-coding and dates show the regions to which the birds had spread at different points in time.

Species Transplants

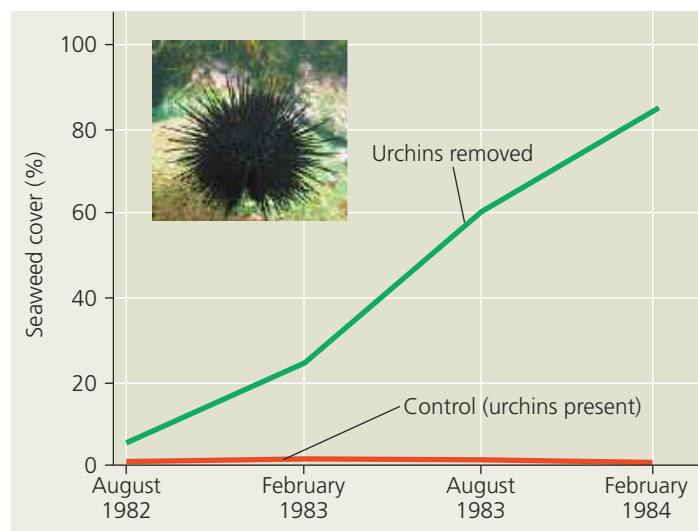
To determine if dispersal is a key factor limiting the distribution of a species, ecologists may observe the results of intentional or accidental transplants of the species to areas where it was previously absent. For a transplant to be successful, some of the organisms must not only survive in the new area but also reproduce there sustainably. If a transplant is successful, then we can conclude that the *potential* range of the species is larger than its *actual* range; in other words, the species *could* live in certain areas where it currently does not.

In some cases, species introduced to new geographic locations have disrupted the communities and ecosystems to which they have been introduced (see Concept 56.1). Consequently, ecologists rarely move species to new regions. Instead, they document the outcome when a species reaches a new region in other ways, such as when a predator is introduced to control a pest species or when a species has been moved accidentally to the area.

Biotic Factors

Our next question is whether biotic factors—other species—limit the distribution of a species. Often, the ability of a species to survive and reproduce is reduced by its interactions with other species, such as predators (organisms that kill their prey) or herbivores (organisms that eat plants or algae). **Figure 52.20** shows how an herbivore, the long-spined sea urchin (*Centrostephanus rodgersii*), has affected the distribution of a food species. In certain marine ecosystems, there is often an inverse relationship between the abundance of sea urchins and seaweeds (multicellular algae, such as kelp). Where urchins that graze on seaweeds and other algae are common, large stands of seaweeds do not become established. As shown in Figure 52.20, Australian

▼ **Figure 52.20 Effects of feeding by sea urchins on seaweed distribution.** Seaweed abundance in areas from which the long-spined sea urchin (*Centrostephanus rodgersii*) had been removed was much higher than seaweed abundance in adjacent control sites from which the urchin was not removed.



researchers tested whether *C. rodgersii* is a biotic factor limiting seaweed distribution. When this urchin was removed from experimental plots, seaweed cover increased dramatically, showing that *C. rodgersii* limited the distribution of seaweeds.

In addition to predation and herbivory, the presence or absence of pollinators, food resources, parasites, pathogens, and competing organisms can act as a biotic limitation on species distribution. Such biotic limitations are common in nature.

Abiotic Factors

The last question in the flowchart in Figure 52.18 considers whether abiotic factors, such as temperature, water, oxygen, salinity, sunlight, or soil, might be limiting a species' distribution. If the physical conditions at a site do not allow a species to survive and reproduce, then the species will not be found there. Throughout this discussion, keep in mind that most abiotic factors vary substantially over space and time. Daily and annual fluctuations of abiotic factors may either blur or accentuate regional distinctions. Furthermore, organisms can avoid some stressful conditions temporarily through behaviors such as dormancy or hibernation (see Concept 40.4).

Temperature

Environmental temperature is an important factor in the distribution of organisms because of its effect on biological processes. Cells may rupture if the water they contain freezes (at temperatures below 0°C), and the proteins of most organisms denature at temperatures above 45°C. Organisms typically function best within a specific range of environmental temperature. Temperatures outside that range may force some animals to expend energy regulating their internal temperature, as mammals and birds do (see Figure 40.16). Extraordinary adaptations